

10. [24th April]

- (i) Find the space complexity involved in the proof of Lemma 1.55 of [Sip].
- (ii) In the NTM N given on page 373, Amelia claims changing *accept* to *reject* and *reject* to *accept* in line 4 decides whether two input NFAs are equivalent. Analyze whether her claim is correct.

9. [23rd April]

- (i) Additional reading (marked AR on the CS205 homepage): go through the polynomial time reduction from TQBF to GG.

8. [22nd April]

- (i) The following example could help in analyzing why an $L \in NP$ does not necessarily immediately imply $\bar{L} \in NP$: Given a polynomial time NTM based decider N for HAMPATH problem, attempt to design a polynomial time NTM based decider (which possibly uses N) for $\overline{\text{HAMPATH}}$. Find reasons for being successful/unsuccessful.
- (ii) By guessing $t(\lg n')$ is $(\lg n')^4$ and substituting the same in the recurrence that was set up in class for the Pratt's algorithm, show the time complexity of Pratt's algorithm is $\text{NTIME}((\lg n)^4)$.
Implement a polynomial time DTM based verifier to determine whether the input positive integer n in binary is a prime.
- (iii) Complete the part of proof left as homework in the coNP lecture note.

7. [17th April]

- (i) Just after three levels of the recurrence tree, how does the fully quantified boolean formula resulting from applying the polynomial time reduction in proving TQBF is PSPACE-hard looks like? Write the formula's structure at the leaf node of the recursion tree.
- (ii) Assuming there is a unique accept ID, following the proof of Cook-Levin theorem, write the unquantified boolean formula for $\phi(I_{start}, I_{accept}, 1)$.
- (iii) Show the FQBFs on the last line of pg 340 of [Sip], the middle of pg 341 of [Sip], and the one written on the board in class are logically equivalent.
- (iv) Given that for any complexity class R , the complexity class $\text{co}R = \{L \mid \bar{L} \in R\}$, prove $\text{PSPACE} = \text{coPSPACE}$.

6. [16th April]

- (i) Assuming Σ of input NFA N is $\{a, b, c\}$, draw the computation tree corresponding to NTM M that decides NOTALLNFA on [Sip] pg 333. Further, in the worst-case, analyze the number of symbols processed along any branch of computation of M on input $\langle N \rangle$, the total number of threads instantiated, and the number of leaves in the computation tree. What about in the best-case?
- (ii) Determine whether the function $n^2(\lg n)^3$ is space constructible.

- (iii) Write the code to generate intermediate ID I' in the Savitch's algorithm.
- (iv) As done in class, formally establish the following: $P \subseteq NP \subseteq PSPACE = NPSpace \subseteq EXP-TIME \subseteq NEXPTIME \subseteq EXPSPACE$.

5. [9th April]

- (i) Exer 8.1, 8.4, 8.8 from [Sip] pg 357-358.
- (ii) Prove whether the following FQBFs belong to TQBF language:
 - (a) $\exists x_1 \forall x_2 \exists x_3 ((x_1 \vee x_2) \wedge (x_2 \vee x_3) \wedge (\bar{x}_2 \vee \bar{x}_3))$, and
 - (b) $\forall x \exists y [(x \vee y) \wedge (\bar{x} \vee \bar{y})]$.
- (iii) Prove the reduction f from 3SAT to 3DM takes polynomial time.
Construct a 3DM instance $f(I)$ corresponding to $(x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_3) \wedge (x_1 \vee \bar{x}_2 \vee x_4)$. By finding whether $f(I) \in 3DM$, determine whether this boolean formula belongs to 3SAT.

4. [8th April]

- (i) Express the length of unary representation of a maximum number in an instance of SUBSET-SUM problem, wherein that instance is obtained by applying the polynomial time reduction from 3SAT, in terms of the size of input 3SAT instance.
- (ii) Exer 7.26 of [Sip]. What if a 2×4 -sized window is used in proving the Cook-Levin's theorem?

3. [3rd April]

- (i) Provide a polynomial time reduction from UHAMPATH to UHAMCYCLE = $\{ \langle G \rangle \mid \text{undirected graph } G \text{ has a Hamiltonian cycle} \}$. Further, without using the transitivity of reductions, give a polynomial time reduction from UHAMPATH to TSP.
- (ii) Let $\phi = (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_4) \wedge (\bar{x}_2 \vee x_3 \vee x_5) \wedge (\bar{x}_3 \vee \bar{x}_4 \vee \bar{x}_5)$. Compute $f(\langle \phi \rangle)$, wherein f is as defined in the polynomial time reduction from 3SAT to HAMPATH. Further, find a HAMPATH in the graph output by f , and determine the corresponding satisfiable truth assignment of ϕ ; if there is no HAMPATH in the graph, argue $\langle \phi \rangle \notin 3SAT$.
- (iii) With ϕ as in the above problem, compute $g(\langle \phi \rangle)$, wherein g is as defined in the polynomial time reduction from 3SAT to SUBSETSUM. Further, determine whether $g(\langle \phi \rangle) \in SUBSETSUM$.

2. [2nd April]

- (i) Give a polynomial reduction from VERTCOVER to CLIQUE.
- (ii) With the two gadgets presented in class for 3SAT to 3COL polynomial time reduction, complete the proof of correctness of $\langle \phi \rangle \in 3SAT \Leftrightarrow f(\langle \phi \rangle) = \langle G \rangle \in 3COL$.
- (iii) Construct graphs' corresponding to (iii)(a) of below problem output by each of the reductions, from 3SAT to CLIQUE, and from 3SAT to 3COL.

1. [1st April]

- (i) Give a polynomial time reduction from 3SAT to SAT, and prove its correctness.

- (ii) While considering all the cases and subcases, as presented in the lecture, complete the proof given on pg 311 of [Sip] for a polynomial time reduction from SAT to 3SAT.
- (iii) Determine whether the following two boolean formulas are satisfiable: (a) $(x_1 \vee x_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee x_2 \vee x_4) \wedge (\bar{x}_2 \vee \bar{x}_3 \vee \bar{x}_5)$, and (b) exer 7.5 on pg 311 of [Sip].